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Self-erecting launcher assembly

Abstract

A self-erecting launcher assembly includes a mobile unit, and a magazine positioned in a receptacle of the mobile unit. The magazine includes a plurality of drones. A rail unit is secured to the mobile unit and moves between a closed position and an elevated position. A platform may be secured to the rail unit and is movable in an upward direction and a downward direction along a length of the rail unit. The platform is movable when the rail unit is in its elevated position. The magazine or the plurality of drones may be positioned on the platform, such that the platform moves the magazine or the plurality of drones from the mobile unit to at least one wellhead. A manipulator arm is secured to the platform and selects a drone of the plurality of drones and delivers the selected drone into a lubricator positioned at the wellhead.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2713909	12/1954	Baker	N/A	N/A
2755863	12/1955	Stansbury et al.	N/A	N/A
3403729	12/1967	Hickey et al.	N/A	N/A
3493061	12/1969	Gyongyosi	N/A	N/A
3669190	12/1971	Sizer et al.	N/A	N/A
3710867	12/1972	Bansbach	N/A	N/A
4678390	12/1986	Bonneton et al.	N/A	N/A
5570992	12/1995	Lemelson	N/A	N/A
6044905	12/1999	Harrison, III	N/A	N/A
6056058	12/1999	Gonzalez	N/A	N/A
6182765	12/2000	Kilgore	N/A	N/A
6269875	12/2000	Harrison, III et al.	N/A	N/A
6454011	12/2001	Schempf et al.	N/A	N/A
6457526	12/2001	Dailey	N/A	N/A
6464011	12/2001	Tubel	N/A	N/A
6474931	12/2001	Austin et al.	N/A	N/A
6488093	12/2001	Moss	N/A	N/A
6659180	12/2002	Moss	N/A	N/A
6719061	12/2003	Muller et al.	N/A	N/A
6808021	12/2003	Zimmerman et al.	N/A	N/A
7018164	12/2005	Anthis et al.	N/A	N/A
7036598	12/2005	Skjærseth et al.	N/A	N/A
7140453	12/2005	Ayling	N/A	N/A
7234525	12/2006	Alves et al.	N/A	N/A
7240742	12/2006	Sewell et al.	N/A	N/A
7273102	12/2006	Sheffield	N/A	N/A
7331394	12/2007	Edwards et al.	N/A	N/A
7588080	12/2008	McCoy	N/A	N/A
7631704	12/2008	Hagemeyer et al.	N/A	N/A
8006765	12/2010	Richards et al.	N/A	N/A
8136585	12/2011	Cherewyk	N/A	N/A

8151882	12/2011	Grigar et al.	N/A	N/A
8317448	12/2011	Hankins et al.	N/A	N/A
8413727	12/2012	Holmes	N/A	N/A
8505632	12/2012	Guerrero et al.	N/A	N/A
8689868	12/2013	Lerche et al.	N/A	N/A
8899322	12/2013	Cresswell et al.	N/A	N/A
8950480	12/2014	Strickland	N/A	N/A
8981957	12/2014	Gano et al.	N/A	N/A
9284168	12/2015	Mau et al.	N/A	N/A
9556676	12/2016	Konduc et al.	N/A	N/A
9617829	12/2016	Dale et al.	N/A	N/A
9797238	12/2016	Frosell et al.	N/A	N/A
9862027	12/2017	Loehken	N/A	N/A
9903192	12/2017	Entchev et al.	N/A	N/A
9951563	12/2017	Beaton et al.	N/A	N/A
9963955	12/2017	Tolman et al.	N/A	N/A
10100612	12/2017	Lisowski et al.	N/A	N/A
10138691	12/2017	Kos et al.	N/A	N/A
10196868	12/2018	Layden	N/A	N/A
10196886	12/2018	Tolman et al.	N/A	N/A
10246952	12/2018	Trydal et al.	N/A	N/A
10301910	12/2018	Whitsitt et al.	N/A	N/A
10323484	12/2018	Liess	N/A	N/A
10358880	12/2018	Metcalf et al.	N/A	N/A
10584552	12/2019	Cannon et al.	N/A	N/A
10605037	12/2019	Eitschberger et al.	N/A	N/A
10612332	12/2019	Sage	N/A	N/A
10739115	12/2019	Loehken et al.	N/A	N/A
10844684	12/2019	Eitschberger	N/A	N/A
10927650	12/2020	Schultz et al.	N/A	N/A
11047189	12/2020	Fernandes et al.	N/A	N/A
11149515	12/2020	Bull et al.	N/A	N/A
11434713	12/2021	Eitschberger et al.	N/A	N/A
11434725	12/2021	Eitschberger et al.	N/A	N/A
11486219	12/2021	Eitschberger	N/A	N/A
12060757	12/2023	Eitschberger	N/A	E21B 23/14
2002/0036101	12/2001	Huhdanmaki et al.	N/A	N/A
2002/0040783	12/2001	Zimmerman et al.	N/A	N/A
2002/0129941	12/2001	Alves et al.	N/A	N/A
2002/0134552	12/2001	Moss	N/A	N/A
2004/0094305	12/2003	Skjærseth et al.	N/A	N/A
2004/0216632	12/2003	Finsterwald	N/A	N/A
2004/0239521	12/2003	Zierolf	N/A	N/A
2005/0011645	12/2004	Aronstam et al.	N/A	N/A
2005/0103526	12/2004	Ayling	N/A	N/A
2005/0217844	12/2004	Edwards et al.	N/A	N/A
2005/0269083	12/2004	Burris, II et al.	N/A	N/A
2006/0054326	12/2005	Alves et al.	N/A	N/A
2008/0223587	12/2007	Cherewyk	N/A	N/A
2009/0211760	12/2008	Richards et al.	N/A	N/A

2009/0255728	12/2008	Spencer et al.	N/A	N/A
2009/0308589	12/2008	Bruins et al.	N/A	N/A
2010/0288496	12/2009	Cherewyk	N/A	N/A
2011/0005777	12/2010	Meff	N/A	N/A
2013/0062055	12/2012	Tolman et al.	N/A	N/A
2013/0228326	12/2012	Griffith et al.	N/A	N/A
2014/0083774	12/2013	Hoult et al.	N/A	N/A
2014/0131035	12/2013	Entchev et al.	N/A	N/A
2014/0218207	12/2013	Gano et al.	N/A	N/A
2014/0360720	12/2013	Corbeil	N/A	N/A
2015/0114626	12/2014	Hatten et al.	N/A	N/A
2015/0136422	12/2014	Rodgers	N/A	N/A
2015/0167410	12/2014	Garber et al.	N/A	N/A
2015/0209954	12/2014	Hokanson	N/A	N/A
2016/0115741	12/2015	Davis	N/A	N/A
2016/0215592	12/2015	Helms et al.	N/A	N/A
2017/0044875	12/2016	Hebebrand et al.	N/A	N/A
2017/0067303	12/2016	Thiemann et al.	N/A	N/A
2017/0067320	12/2016	Zouhair et al.	N/A	N/A
2017/0159379	12/2016	Metcalf et al.	N/A	N/A
2017/0175488	12/2016	Lisowski et al.	N/A	N/A
2017/0204687	12/2016	Yorga et al.	N/A	N/A
2017/0306710	12/2016	Trydal et al.	N/A	N/A
2017/0314372	12/2016	Tolman et al.	N/A	N/A
2017/0370169	12/2016	Genovese et al.	N/A	N/A
2018/0002999	12/2017	Johnson	N/A	N/A
2018/0003038	12/2017	Cherewyk	N/A	N/A
2018/0274311	12/2017	Zsolt	N/A	N/A
2018/0313182	12/2017	Cherewyk et al.	N/A	N/A
2018/0355674	12/2017	Cooper et al.	N/A	N/A
2019/0031307	12/2018	Siersdorfer	N/A	N/A
2019/0186211	12/2018	Gonzalez	N/A	N/A
2019/0218880	12/2018	Cannon et al.	N/A	N/A
2019/0316449	12/2018	Schultz et al.	N/A	N/A
2019/0338606	12/2018	Metcalf et al.	N/A	N/A
2019/0368301	12/2018	Eitschberger et al.	N/A	N/A
2020/0056442	12/2019	Cherewyk et al.	N/A	N/A
2020/0157909	12/2019	Fernandes et al.	N/A	N/A
2020/0190932	12/2019	Eitschberger	N/A	N/A
2020/0308924	12/2019	Pitcher	N/A	N/A
2020/0332618	12/2019	Eitschberger et al.	N/A	N/A
2021/0040809	12/2020	Eitschberger	N/A	N/A
2022/0018196	12/2021	Gregg et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2833722	12/2013	CA	N/A
201546707	12/2009	CN	N/A
204430910	12/2014	CN	N/A
205577894	12/2015	CN	N/A

10582223	12/2016	CN	N/A
10341437	12/2004	DE	N/A
10344523	12/2004	DE	N/A
2952675	12/2014	EP	N/A
2310616	12/2016	EP	N/A
2457081	12/2008	GB	N/A
0133029	12/2000	WO	N/A
0133029	12/2000	WO	N/A
2011150251	12/2010	WO	N/A
2012161854	12/2011	WO	N/A
2015081092	12/2014	WO	N/A
2015081092	12/2014	WO	N/A
2018094220	12/2017	WO	N/A
2019033183	12/2018	WO	N/A
2019229520	12/2018	WO	N/A
2020002383	12/2019	WO	N/A
2020002983	12/2019	WO	N/A
2020035616	12/2019	WO	N/A
2020038848	12/2019	WO	N/A

OTHER PUBLICATIONS

International Searching Authority; International Preliminary Report on Patentability of the International Searching Authority for PCT/EP2020/065180; mailed on Dec. 21, 2021; 7 pages. cited by applicant

International Searching Authority; International Preliminary Report on Patentability of the International Searching Authority for PCT/EP2021/057028; mailed on Sep. 29, 2022; 8 pages. cited by applicant

International Searching Authority; International Search Report and Written Opinion for PCT Appl PCT/EP2020/065180; Oct. 6, 2020; 11 pages. cited by applicant

International Searching Authority; International Search Report and Written Opinion of the International Searching Authority for PCT/EP2021/057028; mailed on Jun. 29, 2021; 11 pages. cited by applicant

Kort Industries, Self Elevating Drilling System, 9 pgs., Apr. 17, 2020, <https://www.kortindustries.com/seds/>. cited by applicant

United States Patent and Trademark Office, Final Office Action of U.S. Appl. No. 16/423,230, dated Nov. 4, 2019, 14 pages. cited by applicant

United States Patent and Trademark Office, Non-Final Office Action of U.S. Appl. No. 16/423,230, dated Aug. 27, 2019, 16 pages. cited by applicant

United States Patent and Trademark Office, Non-Final Office Action of U.S. Appl. No. 16/788,107, dated Apr. 6, 2020, 15 pages. cited by applicant

United States Patent and Trademark Office, Notice of Allowance for U.S. Appl. No. 16/423,230, dated Nov. 27, 2019, 7 pages. cited by applicant

United States Patent and Trademark Office, Notice of Allowance for U.S. Appl. No. 16/788,107, dated Jul. 30, 2020, 9 pages. cited by applicant

United States Patent and Trademark Office; Non-Final Office Action for U.S. Appl. No. 16/919,473; dated Feb. 8, 2022; 12 pages. cited by applicant

United States Patent and Trademark Office; Non-Final Office Action for U.S. Appl. No. 17/072,067; dated Mar. 31, 2022; 15 pages. cited by applicant

United States Patent and Trademark Office; Non-Final Office Action for U.S. Appl. No. 17/608,173; dated Mar. 29, 2022; 5 pages. cited by applicant

United States Patent and Trademark Office; Notice of Allowance for U.S. Appl. No. 16/919,473; dated Jun. 14, 2022; 8 pages. cited by applicant

United States Patent and Trademark Office; Notice of Allowance for U.S. Appl. No. 17/072,067; dated Aug. 12, 2022; 10 pages. cited by applicant

United States Patent and Trademark Office; Notice of Allowance for U.S. Appl. No. 17/608,173; dated Jun. 20, 2022; 8 pages. cited by applicant

Entchev et al., "Autonomous Perforating System for Multizone Completions," SPE 147296, Prepared for Presentation at Society of Petroleum Engineers (SPE) Annual Technical Conference and Exhibition held Oct. 30, 2011-Nov. 2, 2011, 7 pgs. cited by applicant

International Searching Authority, International Search Report and Written Opinion of International App. No. PCT/EP2019/063966, mailed Aug. 30, 2019, 10 pages. cited by applicant

International Searching Authority, International Search Report and Written Opinion for PCT App. No. PCT/IB2019/000526; Sep. 25, 2019, 17 pgs. cited by applicant

International Searching Authority, The International Search Report and Written Opinion of International App. No. PCT/IB2019/000537, mailed Sep. 25, 2019, 18 pgs. cited by applicant

International Searching Authority; Communication Relating to the Results of the Partial International Search for PCT/EP2020/070291; Oct. 20, 2020; 8 pages. cited by applicant

International Searching Authority; International Preliminary Report on Patentability for International Application No. PCT/IB2019/000537; dated Dec. 10, 2020; 11 pages. cited by applicant

International Searching Authority; International Preliminary Report on Patentability for International Application No. PCT/IB2019/000526; dated Dec. 10, 2020; 10 pages. cited by applicant

International Searching Authority; International Preliminary Report on Patentability International Application No. PCT/EP2019/063966; dated Dec. 10, 2020; 7 pages. cited by applicant

International Searching Authority; International Search Report and Written Opinion for PCT App. No. PCT/EP2019/066919; Sep. 10, 2019; 11 pages. cited by applicant

International Searching Authority; International Search Report and Written Opinion for PCT App. No. PCT/EP2019/072032; Nov. 15, 2019; 13 pages. cited by applicant

International Searching Authority; International Search Report and Written Opinion for PCT App. No. PCT/EP2019/072064; Nov. 20, 2019; 15 pages. cited by applicant

International Searching Authority; International Search Report and Written Opinion of the International Searching Authority for PCT/EP2020/070291; Dec. 15, 2020; 14 pages. cited by applicant

United States Patent and Trademark Office, Final Office Action of U.S. Appl. No. 16/542,890, dated May 12, 2020, 16 pgs. cited by applicant

United States Patent and Trademark Office, Non-Final Office Action of U.S. Appl. No. 16/423,230, which is in the same family as U.S. Appl. No. 16/788,107, dated Aug. 27, 2019, 16 pgs. cited by applicant

United States Patent and Trademark Office, Non-final Office Action of U.S. Appl. No. 16/451,440, dated Oct. 24, 2019, 22 pgs. cited by applicant

United States Patent and Trademark Office, Notice of Allowance for U.S. Appl. No. 16/788,107, mailed Jul. 30, 2020, 9 pgs. cited by applicant

United States Patent and Trademark Office; Final Office Action for U.S. Appl. No. 16/451,440; issued Feb. 7, 2020; 11 pages. cited by applicant

United States Patent and Trademark Office; Non-Final Office Action for U.S. Appl. No. 16/542,890; issued Nov. 4, 2019; 16 pages. cited by applicant

United States Patent and Trademark Office; Non-Final Office Action for U.S. Appl. No. 16/542,890; Sep. 30, 2020; 17 pages. cited by applicant

United States Patent and Trademark Office; Notice of Allowance for U.S. Appl. No. 16/451,440;

issued Jun. 5, 2020; 8 pages. cited by applicant

USPTO, Final Office Action of U.S. Appl. No. 16/423,230, dated Nov. 4, 2019, 14 pgs. cited by applicant

USPTO, Office Action of U.S. Appl. No. 16/788,107, dated Apr. 6, 2020, 15 pgs. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/911,456 filed Sep. 14, 2022, which is a national stage application of and claims priority to Patent Cooperation Treaty (PCT) Application No. PCT/EP2021/057028 filed Mar. 18, 2021, which claims the benefit of U.S. Provisional Application No. 62/991,125 filed Mar. 18, 2020, the entire contents of each which are incorporated herein by reference in their entireties.

BACKGROUND

(1) Oil and gas reserves are accessed using various drilling and completion techniques. The drilling techniques require preparation of a drilling site by the formation of a wellbore **50**, as illustrated in FIG. **1**. A wellbore **50** is a narrow shaft drilled in the ground, vertically and/or horizontally as well as angles therebetween. A wellbore **50** can include a substantially vertical portion and a substantially horizontal portion. The vertical portion of a typical wellbore **50** may be over a mile in depth, while the horizontal portion may be several miles in length due to longer laterals and advancements in horizontal drilling.

(2) A tool **31** or tool string **31'** is typically introduced into a wellbore **50** by attaching a lubricator **23** to a blowout preventer **25** at the wellhead **30** of a well casing. The lubricator **23** is a series of large diameter tubular members assembled on top of wellhead **30** and may include a grease injection tube and/or stuffing box **27** through which a wireline **24** for suspending the tool **31** is passed. The lubricator **23** is typically long, heavy and difficult to manipulate in the rig **26**. It may also be difficult to make the required connections to a lubricator **23**. After the lubricator **23** and stuffing box **27** have been assembled, the lubricator **23** is hoisted into position on the blowout preventer **25** and secured thereto. Pressure between the wellbore **50** and lubricator **23** is equalized by valves around the blowout preventer **25**. The blowout preventer **25** is then opened allowing access to the borehole. After the blowout preventer **25** has been opened, the tool **31** can be lowered into the wellbore **50** by a wireline **24** with the grease injection tube or stuffing box **27** providing a seal around the wireline **24** as the tool **31** is lowered.

(3) Once the tool has served its desired purpose in the wellbore **50**, the tool **31** is extracted from the wellbore **50** by drawing it up to a position within the lubricator **23**, closing the blowout preventer **25**, venting the lubricator **23**, and removing the tool **31**. When no more tools will be deployed in the wellbore, the lubricator **23** may be removed from the blowout preventer **25** and lowered to a position where it can be subsequently disassembled into its individual components. It will be appreciated from the foregoing description that there are a number of difficulties in such an operation, including knowing when the tool **31** has been fully withdrawn into the lubricator **23**, not pulling the wireline **24** so taut against the stuffing box **27** that there is a possibility of the wireline **24** being broken with the result being the tool **31** falling downhole before the blowout preventer **25** can be closed, and closing the blowout preventer **25** on the tool **31** before it is fully withdrawn into the lubricator **23**. Of course, handling the tool **31** during the extraction process is equally as difficult as handling it during the insertion process.

(4) A wireline, electric line, or e-line **24** is cabling technology used to lower and retrieve tools **31** into and out of the wellbore **50** for the purpose of delivering an explosive charge, evaluation of the wellbore **50**, or other completion-related or closure-related tasks. The equipment/devices disposed in the wellbore **50** are often generically referred to as downhole tools and examples of such tools **31** are perforating guns, puncher guns, logging tools, jet cutters, plugs, frac plugs, bridge plugs, setting tools, self-setting bridge plugs, self-setting frac plugs, mapping/positioning/orientating tools, bailer/dump bailer tools, and ballistic tools. Such downhole tools **31** are typically attached to the wireline **24**, fed through or run inside the casing or tubing, and are lowered into the wellbore **50**. Other methods include tubing conveyed (i.e., TCP for perforating) or coil tubing conveyance. The speed of unwinding the wireline cable **24** and winding the wireline cable **24** back up is limited based on a speed of the wireline equipment rig **26** and forces on the wireline cable **24** itself (e.g., friction within the well). Because of these limitations, it typically takes several hours for the wireline cable **24** and the attached tool **31** or tool-string **31'** to be lowered into the wellbore **50** and another several hours for the wireline cable **24** to be wound back up and the toolstring **31'** retrieved. When detonating explosives, the wireline cable **24** will be used to position a downhole tool **31** or toolstring **31'** into the wellbore **50**.

(5) This type of deployment process requires the selection of a downhole tool **31**, the attachment of that tool **31** or a combination of tools in a toolstring **31'** to the wireline **24**, and in some instances, the removal of the downhole tool(s) **31** from the wellbore **50**. When an operator needs to deploy additional downhole tools **31** into the wellbore **50**, which may be the same as or different from previously-deployed tool(s), the operator must first retract/retrieve the wireline **24** from the wellbore **50** and then attach the wireline **24** to the additional downhole tool(s) **20**. That is, no practical means exists for disposing more than one wireline **24** into a wellbore **50** during typical operations. This completion process requires multiple steps, a significant array of equipment, and can be time consuming and costly. Furthermore, equipment lodged in the wellbore will typically result in complication, delay, additional human resource time, equipment cost and, often, exorbitant expense to operations.

(6) The various drilling and completion operations requiring deployment of various downhole tools **31**, as well as the changing between different types of tools being deployed, currently require direct human interaction with the wireline **24**, the tools **31** on the wireline **24**, and the feeding of tools/wireline into the equipment attached to the wellhead **30**. Wellhead **30** is a general term used to describe the pressure-containing component at the surface of an oil well that provides the interface for drilling, completion, and testing of all subsurface operation phases. Being pressurized and the pressurization subject to an unknown level of variability, in addition to the substantial amount of shifting equipment adjacent the wellhead **30**, the area around the wellhead **30** is referred to as a 'red zone'. That is, the dangers inherent in drilling and completion operations are focused in the area within a few yards or tens of yards around the wellhead **30**. During operations, only trained personnel are permitted within a certain distance of the wellhead **30** and those personnel must be properly protected. Even then, the activities of attaching and detaching tools **31** from a wireline **24**, disposing a wireline **24** and attached toolstring **31'** into the wellbore **50** and retrieving a wireline **24** and the attached toolstring **31'** from the wellbore **50**, are inherently difficult, dirty and dangerous.

(7) In view of the disadvantages associated with currently available devices and methods for well completion, there is a need for a device and method that increases the efficiency of the completion processes. There is a further need for a device and method that reduces the steps, time to achieve steps, time between steps and associated costs and equipment for well completion processes. There is a further need for a system and method that reduces the delay between drilling of a wellbore and production of oil or gas from the wellbore. In light of the dangers of disposing and retrieving tools from a wellbore, there is also a need to reduce or eliminate the number of persons in the red zone adjacent the wellhead, especially during particularly risk prone activities.

BRIEF SUMMARY

(8) Embodiments of the disclosure are associated with a self-erecting launcher assembly. The launcher assembly includes a mobile unit, and a magazine positioned in a receptacle of the mobile unit. According to an aspect, the magazine includes a plurality of drones. A rail unit is secured to the mobile unit **400** and is configured to move between a closed position and an elevated position. According to an aspect, a platform **600** is secured to the rail unit and is movable in an upward direction and a downward direction along a length of the rail unit. The platform **600** may be movable sideways. is side-by-side direction in order to orient a manipulator arm. The platform is movable when the rail unit is in its elevated position. According to an aspect, the magazine or the plurality of drones may be positioned on the platform, such that the platform moves the magazine or the plurality of drones from the mobile unit to at least one wellhead.

(9) Embodiments of the disclosure may be further directed to a wellbore deployment system including a plurality of self-erecting launcher assembly. Each self-erecting launcher assembly of the plurality of self-erecting launcher assembly may be configured substantially as described hereinabove. According to an aspect, each self-erecting launcher assembly of the plurality of self-erecting launcher assembly includes mobile unit having a receptacle, and a magazine including a plurality of drones and positioned in the receptacle. A rail unit may be secured to the mobile unit. According to an aspect, the rail unit is movable between a closed position and an elevated position and a platform is secured to the rail unit. The platform may be movable in an upward direction and a downward direction along a length of the rail unit, such that the platform moves the magazine or the plurality of drones from the mobile unit to at least one wellhead. The wellbore deployment system further includes a control unit configured to manage functionality and operation of the plurality of self-erecting launcher assembly.

(10) Further embodiments of the disclosure may be associated with a wellbore deployment system including a self-erecting launcher assembly. The self-erecting launcher assembly includes a mobile unit including a receptacle. A magazine may be positioned in the receptacle. According to an aspect, the magazine includes a plurality of drones. A rail unit is secured to the mobile unit, and is movable between a closed position and an elevated position. According to an aspect, a platform is secured to the rail unit and is movable in an upward direction and a downward direction along the length of the rail unit. A location adjuster is configured to move the self-erecting launcher assembly from a first wellhead to additional wellheads at a well site.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.
- (2) FIG. 1 illustrates a wellbore deployment system, according to the prior art;
- (3) FIG. 2 illustrates a drone, according to the prior art;
- (4) FIG. 3A illustrates a perspective view of a self-erecting launcher assembly, according to an aspect;
- (5) FIG. 3B illustrates a side view of the self-erecting launcher assembly of FIG. 3A;
- (6) FIG. 4 illustrates a side view of a mobile unit including a magazine loading bay positioned in at least one side wall of the mobile unit, according to an aspect;
- (7) FIG. 5 illustrates a side view of a mobile unit including a magazine loading bay positioned in the floor of the mobile unit, according to an aspect;
- (8) FIG. 6 illustrates the movement of a drone within a receptacle of a mobile unit, according to an aspect;
- (9) FIG. 7 illustrates a top view of a self-erecting launcher assembly, according to an aspect;
- (10) FIG. 8 illustrates a top view of a mobile unit positioned on a rail unit, according to an aspect;

- (11) FIG. 9 illustrates a top view of a mobile unit of a self-erecting launcher assembly, configured to service a plurality of wellheads, according to an aspect;
- (12) FIG. 10 illustrates a top view of a mobile unit **400** of a self-erecting launcher assembly **300** positioned on a rotational unit, according to an aspect;
- (13) FIG. 11 illustrates a side, perspective view of a rail unit in an elevated position and a platform secured to the rail unit and positioned to deploy a drone in a wellhead, according to an aspect;
- (14) FIG. 12 illustrates a top view of the platform of FIG. 11;
- (15) FIG. 13 illustrates a perspective view of the platform of FIG. 11, according to an aspect;
- (16) FIG. 14 illustrates a platform slidably positioned on a track rail, according to an aspect;
- (17) FIG. 15 illustrates a perspective view of a control unit of a self-erecting launcher assembly, according to an aspect; and
- (18) FIG. 16 illustrates a perspective view of a device for delivering a magazine to a mobile unit, according to an aspect.
- (19) Various features, aspects, and advantages of the exemplary embodiments will become more apparent from the following detailed description, along with the accompanying drawings in which like numerals represent like components throughout the figures and detailed description. The various described features are not necessarily drawn to scale in the drawings but are drawn to emphasize specific features relevant to some embodiments.
- (20) The headings used herein are for organizational purposes only and are not meant to limit the scope of the disclosure or the claims. To facilitate understanding, reference numerals have been used, where possible, to designate like elements common to the figures.

DETAILED DESCRIPTION

- (21) Reference will now be made in detail to various embodiments. Each example is provided by way of explanation and is not meant as a limitation and does not constitute a definition of all possible embodiments.
- (22) Embodiments of the disclosure are associated with, among other things, a self-erecting launcher assembly **300**/a drone delivery system/an autonomous tool delivery system. In an exemplary embodiment, the autonomous tool may be, without limitation, a drone. For purposes of this disclosure, a “drone” is a self-contained, autonomous or semi-autonomous vehicle for downhole delivery of a wellbore tool. For purposes of this disclosure and without limitation, “autonomous” means without a physical connection or manual control (such as a wireline). The assembly presented is versatile and facilitates the delivery of autonomous tools to a single well or multiple wells and facilitates a simplified and accurate mode of delivering such drones to a wellhead **950**.
- (23) For purposes of illustrating features of the embodiments, embodiments of the disclosure will now be introduced in reference to the figures. Those skilled in the art will recognize that this example is illustrative and not limiting and is provided purely for explanatory purposes.
- (24) This application incorporates by reference each of the following pending patent applications in their entireties: U.S. Provisional Patent Application No. 62/842,329, filed May 2, 2019; U.S. Provisional Patent Application No. 62/841,382, filed May 1, 2019; International Patent Application No. PCT/IB2019/000526, filed Apr. 12, 2019; U.S. Provisional Patent Application No. 62/831,215, filed Apr. 9, 2019; International Patent Application No. PCT/IB2019/000530, filed Mar. 29, 2019; U.S. Provisional Patent Application No. 62/832,737, filed Mar. 26, 2019; International Patent Application No. PCT/IB2019/000537, filed Mar. 18, 2019; U.S. Provisional Patent Application No. 62/816,649, filed Mar. 11, 2019; U.S. Provisional Patent Application No. 62/720,638, filed Aug. 21, 2018; U.S. Provisional Patent Application No. 62/765,185, filed Aug. 16, 2016; U.S. Provisional Patent Application No. 62/719,816, filed Aug. 20, 2018; U.S. Provisional Patent Application No. 62/690,314, filed Jun. 26, 2018; U.S. Provisional Patent Application No. 62/678,654, filed May 31, 2018; and U.S. Provisional Patent Application No. 62/678,636, filed May 31, 2018.

(25) In general, the embodiments of the disclosure concern the use of one or more drones in drilling and, especially, well completion operations. As used herein, the term “drone” refers to a downhole tool or toolstring not connected to a physical wire/cable, i.e., the term “drone” refers generally to an untethered downhole tool. Drones are configured for deployment into and use in a wellbore. The drone may be configured to move at pump speed or flow rate speed (i.e., the speed at which fluid is pumped into the wellbore).

(26) With reference to FIG. 2, an exemplary embodiment of a drone **10** is shown. As described herein, the drone **10** may include a wellbore tool and may be launched autonomously or semi-autonomously into the wellbore **50**. The wellbore tools may include, for example and without limitation, a perforating gun, puncher gun, logging tool, jet cutter, plug, frac plug, bridge plug, setting tool, self-setting bridge plug, self-setting frac plug, mapping/positioning/orientating tool, bailer/dump bailer tool and ballistic tool. The wellbore tool drones **10** are deployed in the wellbore **50**, without requiring a wireline assembly.

(27) The exemplary drone **10** shown in FIG. 2 is in the configuration of a downhole tool often referred to as a perforating gun. This perforating gun drone includes a body portion **52** having a front end **54** and a rear end **56**. A head portion **58** extends from the front end **54** of the body portion **52** and a tail portion **60** extends from the rear end **56** of the body portion **52** in a direction opposite the head portion **58**. The body portion **52** includes a plurality of shaped charge apertures **74** and open apertures **64**. Each of the plurality of shaped charge apertures **74** are configured for receiving and retaining a shaped charge **62**.

(28) In the exemplary perforating gun drone embodiment, the body portion **52** is a unitary structure that may be formed from an injection-molded material, as are the head portion **58** and the tail portion **60**. In other embodiments, the body portion **52**, the head portion **58** and the tail portion **60** may constitute modular components or connections. As shown in FIG. 2, each of the body portion **52**, the head portion **58** and the tail portion **60** is substantially cylindrically-shaped and may include fins **70**. Each of these features is configured with regard to its travel as a drone **10** into the wellbore **50**.

(29) Turning now to FIG. 3A and FIG. 3B, a self-erecting launcher assembly **300** is illustrated. The self-erecting launcher assembly **300** maybe configured to convey a drone **10** into a wellbore. According to an aspect and as described in further detail hereinbelow, the self-erecting launcher assembly **300** is configured to convey a plurality of drones into a plurality of wellbores. The self-erecting launcher assembly **300** includes a mobile unit **400**, a rail unit **500** coupled to the mobile unit **400**, and a platform **600** secured to the rail unit **500** and movable in an upward direction and a downward direction along a length of the rail unit **500**.

(30) As seen in FIG. 4, the mobile unit **400** includes a receptacle **401**. The mobile unit **400** further includes at least one side wall **402** having an upper portion **403** and a lower portion **404**, a roof **405** connected to the upper portion of the at least one side wall, and a floor **406** connected to the lower portion **404** of the at least one side wall **402**. The receptacle **401** may be bounded by the at least one side wall **402**, the floor **406** and the roof **405**. According to an aspect, the mobile unit **400** is configured as a truck or a portion of a truck, such as a flatbed, a c-bed or a box unit of a trailer. The mobile unit **400** may include a stabilizer unit **440**, described in further detail hereinbelow, to secure the mobile unit **400** in place. Alternatively, or in combination with the stabilizer unit **440**, the mobile unit **400** may include a plurality of wheels **410** that allows the mobile unit **400** to move between different locations, such as between a plurality of wellheads **950**.

(31) The mobile unit **400** may include a hydraulic unit **450** to power one or more components of the self-erecting launcher assembly **300**. The hydraulic unit **450** may be a self-contained system. According to an aspect the hydraulic unit **450** includes a motor, a receptacle including hydraulic fluid, and a pump. One or more filters are provided to prevent contaminants such as dust and dirt from the well site from entering the receptacle of the hydraulic unit **450**, thereby keeping the hydraulic fluid clean. The hydraulic unit **450** applies the hydraulic pressure in order to power the

components of the self-erecting launcher assembly **300**, such as hydraulic pistons in the telescoping elements **335** and any other mechanical elements required to erect the launcher assembly including the side rails **330** from the horizontal position into the vertical position.

(32) According to an aspect and as illustrated in FIG. 4 and FIG. 5, the mobile unit **400** includes a magazine loading bay **420**. The magazine loading bay **420** may be configured as an opening **421** that serves as a conduit between an area outside of the receptacle **401** and an area within the receptacle **401**. The magazine loading bay **420** may be in communication with the receptacle **401**, such that the receptacle **401** is accessible by a user through the magazine loading bay **420**.

(33) As illustrated in FIG. 4 the magazine loading bay **420** may be positioned in the at least one side wall **402** of the mobile unit **400**. When the magazine loading bay **420** is configured as an opening **421**, a door or covering member (not shown) may be provided over the opening **421**. The door may be movable between open and closed positions to provide access to and to limit access to the receptacle **401**. According to an aspect, the door may be connected to the side wall of the mobile unit **400** by hinges, such that the door can swings between the open and closed positions. The door may be configured as a drop door connected to the at least one side wall **402** by two or more hinge elements positioned along a bottom edge of the door.

(34) Alternatively, and as illustrated in FIG. 5, the magazine loading bay **420** may be positioned in the floor **406** of the mobile unit **400**. In this configuration, the magazine loading bay **420** may include a hatchway door **422**. Alternatively, the door may be a sliding door. When the door is positioned in the floor **406** of the mobile unit **400**, a lift mechanism (not shown) may be employed to move a magazine **100** into and out of the receptacle **401** of the mobile unit **400**.

(35) According to an aspect and as illustrated in FIG. 6, the mobile unit **400** further comprises a conveyor system **430**. The conveyor system **430** is arranged in the receptacle **401** and is configured to move an object, such as a magazine or a drone **10** from the magazine loading bay **420** to a magazine retrieval zone **416** (see FIG. 7). According to an aspect, the conveyor system **430** includes a conveyor belts operable to transport a magazine **100** or a drone (positioned within or external to a magazine **100**). The conveyor belts of the conveyor system **430** may be configured to transport a plurality of the magazines **100** and/or drones simultaneously. According to an aspect, the conveyor system **430** includes rollers to help reduce the friction between the magazines **100** or drones and the conveyor belt. As illustrated in FIG. 6, the conveyor system **430** may be arranged on a path where the magazines **100** or drones travel along the path independent of each another. One or more conveyor platforms (not shown) may be positioned on the conveyor belt in order to serve as a designated surface upon which the magazines **100** and drones can be positioned. The conveyor system **430** and empty or rejected magazines **100** are moved to the back of the mobile unit **400** via the same loop. An alternate magazine **100** return technique may include lowering a magazine **100** through a hatch door or opening **407** formed in the floor **406** of the mobile unit **400**.

(36) When the magazines **100** are loaded into the receptacle **401** of the mobile unit **400** through the magazine loading bay **420**, they are moved internally via, for example, by the conveyor system **430** to the front portion **412** of the mobile unit **400**. As illustrated in FIG. 7, the magazine retrieval zone **416** may be positioned at the front portion **412** of the mobile unit **400**.

(37) Once a magazine **100** is in the front portion **412** of the mobile, drones are charged (electrically) within the magazine **100** from a power source. Because the drone may include an explosive component, in order to reduce or inhibit the potential impact of inadvertent detonation of the explosive component of the drone **10**, one or more surfaces of the mobile unit **400** may include a metal foam **433**. According to an aspect, the front of the receptacle **401** is specially shielded against internal blast by the metal foam **433**. In some embodiments, the metal foam **433** includes at least one of aluminum, steel, iron, or combinations thereof. The metal foam **433** may line an inner surface of the upper portion **403** of the mobile unit **400**. According to an aspect, the metal foam **433** may line the inner surface of at least one of the side walls, the roof **405** and the floor **406** of the mobile unit **400**. The metal foam **433** may be composed of various metal alloys. In some

embodiments, the metal foam **433** is a porous irregular structure and may be formed from various methods, including gas injection within a metallic structure, powder metallurgy, casting, metallic deposition, sputter deposition, and/or heat treatment of aluminum powder. The metal foam **433** may be bonded together with sheet metal composed of various metal alloys, such as steel. The metal foam **433** prevents and/or limits ballistic transfer from one drone/autonomous wellbore device to another drone or autonomous wellbore device in the event of detonation of an explosive component of the drone or autonomous wellbore device.

(38) As described herein, the magazines **100** may contain one or more drones for delivery into the wellbore. For purposes of the disclosure, exemplary embodiments of a launcher assembly and delivery system are described for brevity for use with a drone, although the disclosure is not limited thereto and the launcher assembly/delivery system is not limited to use with any particular wellbore tool(s).

(39) The magazine contemplated for use with the self-erecting launcher assembly **300** may be configured substantially as the magazine described in US Publication No. US 2020/0332618, which is commonly-owned and assigned to DynaEnergetics Europe GmbH, the disclosure of which is incorporated herein by references in its entirety.

(40) According to an aspect, the magazine **100** includes a magazine frame serving the function of holding the plurality of drones. The magazine frame may be divided into multiple sections. For example, first section of the magazine frame may hold a first group of drones and second section of the magazine **100** frame may hold a second group of drones. In addition, other multi-segment magazine **100** frames may hold other groups of drones. Each group of drones may, whether occupying a single magazine **100** or multiple magazines, comprise a single tool. That is, tools having different functions may be selected from one or more magazines and dropped into the wellbore **50** in a predetermined and useful order. Alternatively, different groups of drones may be the same tool but with configuration details varying from group to group. Tools with a particular configuration may be placed in the wellbore **50** in a predetermined and useful order. In another embodiment, a magazine **100** may be loaded with drones of different types or configurations in the order in which it is desired to drop the drones into the wellbore. In this case, switching magazines is unnecessary except to the extent that a magazine **100** has been exhausted of drones **10**.

(41) The magazine **100** may also include at least one magazine transceiver configured to communicate with the drone **10**. According to an embodiment, the at least one magazine transceiver is received within each of the magazine chambers. Alternatively, a single magazine transceiver is provided with each magazine **100** and relays information regarding the drones **10**. The magazine transceiver may receive information transmitted from a communication with a drone transceiver included in the drone **10**. According to an aspect, the drone transceiver may be as simple as a radio-frequency identification (RFID) tag, an optical marker such as a QR code or bar code or a data matrix code. It is contemplated that the magazine transceiver may communicate with one or more transceivers included in the drone **10**.

(42) In an embodiment, the magazine transceiver receives information from a plurality of sensors. The sensors may be configured to perform at least one of a plurality of functions. According to an aspect, the sensors are configured to detect the presence of the drone **10** in the magazine chamber. If the sensor in one of the magazine chambers determines that no drone **10** is present, the release element corresponding with that magazine chamber will remain in its closed position.

(43) According to an aspect, the sensors may distinguish between different types of drone **10**. This may be particularly important when selecting the type of drone **10** that should be dispensed from the magazine **100**. The sensors may be configured to measure a voltage level of a battery housed within the drone **10**.

(44) In an embodiment, the magazine **100** is configured to perform one or more self-tests in response to a command from a control unit **700** (see FIG. 15). The control unit **700** may be electrically connected to one or more of the magazines **100**, the magazine chambers and the drone

10 by one of a wireless local area network (LAN) connection, a wireless connection such as through a Bluetooth, and a plug-in adapter connection. According to an aspect, each of the magazine chambers is automatically locked in place based on the information received by the magazine transceiver or the results of the one or more tests.

(45) The magazine chambers may also include one or more safety device actuators. According to an aspect, the one or more safety device actuators may include a ballistic limiter **111** configured to prevent or limit ballistic transfer from a first drone to a second drone positioned in the magazine **100** in the event of detonation of an explosive component of the first drone. The ballistic limiter **111** may include metal foam, configured substantially as described hereinabove.

(46) As illustrated in FIG. 7, at least one stabilizer unit **440** may be positioned on a lower portion **404** of the mobile unit **400**. The at least one stabilizer unit **440** may help withstand the weight and operation of the magazines **100**, and other components described in further detail hereinbelow (such as rails, a platform **600**, and motor used with the self-erecting launcher assembly **300**). The at least one stabilizer unit **440** may include a generally U-shaped bar along the floor **406** of the mobile unit **400** connected to plurality of wheels **410**. According to an aspect, a plurality of stabilizer units may be provided, with a first stabilizer unit being positioned at the front portion **412** of the mobile unit **400** and a second stabilizer unit being positioned at a rear portion of the mobile unit **400**.

(47) According to an aspect, the at least one stabilizer unit **440** includes at least one stabilizer bar **441** extending vertically from the lower portion of the at least one side wall to a ground surface. A pair of spaced-apart vertically extending stabilizer bars may be provided, each stabilizer bar being spaced apart from another stabilizer bar. The vertically extending stabilizer bars may each extend between the lower portion of the at least one side wall to the ground surface. According to an aspect, a horizontal stabilizer bar/cross bar (not shown) may extend along the floor **406** portion of the mobile unit **400**, with each end of the horizontal stabilizer bar being connected to a vertically extending stabilizer bar that extends to the ground surface.

(48) While FIG. 8 illustrates the self-erecting launcher assembly **300** being positioned to service a single wellbore **50** or a single wellhead **950**, it is contemplated that the mobile unit **400** may be movable between a plurality of wellheads **950**.

(49) The self-erecting launcher assembly **300** may be configured to may serve multiple wells. According to an aspect, and as illustrated in FIG. 3, the mobile unit **400** may include a plurality of wheels **410** that allows the mobile unit **400** to move between different locations, e.g., by towing, such as between a plurality of wellheads **950**.

(50) Alternatively, and as illustrated in FIG. 9, the mobile unit **400** is positioned to travel on a track rail **800** secured to a ground surface, such that the mobile unit **400** can access and service multiple wellheads **950**. The track rail **800** may include a track section including one or more moving elements arranged to travel along the track section. According to an aspect, the track rail **800** includes a plurality of track segments that are modular. The track rail segments may be connected, and quickly and easily disconnected or separated from one another. According to an aspect, the track rail segments may be connected to form a long track rail so that the plurality of wellheads **950** may be accessible by the mobile unit **400**. According to an aspect, the track segments may be mounted on an elevated support (not shown) so that the track segments are spaced apart from the ground surface.

(51) According to an aspect, a location sensor or detector (not shown) may be positioned on the track rail segments. The location sensor may be positioned on the mobile unit **400**. The location sensor may be disposed on the track rail segments or on the mobile unit **400** such that a location of the mobile unit **400** unit may be easily determined.

(52) In an embodiment and as illustrated in FIG. 10, the mobile unit **400** can be installed on a rotational unit **850** that allows the self-erecting launcher assembly **300** to service multiple wellheads **950**. The rotational unit **850** includes a rotatable base **851** upon which the mobile unit **400** is positioned. According to an aspect, the rotatable base **851** is rotatable 360-degrees so that the

front portion **412** of the mobile unit **400** can be positioned so that the magazine retrieval zone **416** is adjacent a wellhead **950**. As illustrated in FIG. **10**, the mobile unit **400** may rotate between a first wellhead **950** and a second wellhead **950**.

(53) In aspects/alternative arrangements of the exemplary embodiments, the mobile unit **400** may be elevated directly on a scissor lift (not shown) or the like for elevating the mobile unit **400** to a portion or full height of the wellhead **950** or lubricator **955**. This may, for example, reduce or eliminate the time required for a platform **600**, including a drone delivery arm, to travel up and down between the receptacle **401** and the top of the wellhead **950** or lubricator **955**.

(54) As illustrated in FIG. **11** and FIG. **12**, the self-erecting launcher assembly **300** may include a pair of spaced apart side rails **330** that are movable between a closed (non-erected) position and an open (erected or open) position. The pair of spaced apart side rails **330** may be connected to the mobile unit **400** by a hinge connection. Each side rail of the pair of spaced apart side rails **330** includes a connected end **331** and a free end **333**. According to an aspect, a telescoping element **335** extends between the connected end **331** and the free end **333**, and the mobile unit **400**.

According to an aspect, the telescoping element **335** includes a pair of telescoping arms (not shown). A first telescoping arm **337** of the pair of telescoping arms may be configured to move a first rail **330a** of the pair of spaced apart side rails **330**, and a second telescoping arm (not shown) of the pair of telescoping arms **336** may be configured to move a second rail **330b** of the pair of spaced apart side rails **330**. The first telescoping arm **337** is connected to the first rail **330a** at an intermediate position between the first end and the second end of the first rail **330a**. The second telescoping arm is connected to the second rail **330b** at an intermediate position between the first end and the second end of the second rail **330b**. The rails may be raised from the mobile unit **400**, via the first telescoping arm **337** and the second telescoping arm.

(55) When in their fully erected positioned, the first and second side rails **330a**, **330b** may be positioned at about a 90-degree angle. This may help to enhance the stability of the self-erecting launcher assembly **300** so that it remains secure to the ground surface. According to an aspect, the pair of spaced apart side rails **330** may be detachable from the mobile unit **400** (not shown). In its detached configuration, the pair of spaced apart side rails **330** may move between multiple wellheads **950**.

(56) A platform **600** may be positioned between and slidably connected to each rail of the pair of spaced apart side rails **330**. According to an aspect, the platform **600** includes a right side connector **610** connected to the first rail and a left side connector **620** connected to the second rail **330b**. The right side connector **610** includes a protrusion (not shown) that is received in a groove formed in the first rail **330a** and the left side connector **620** includes a protrusion that is received in a groove formed in the second rail **330b**. The right and left side connectors **610**, **620** help to facilitate movement of the platform **600** along the rail unit **500**, which helps to move drones from the mobile unit **400** to the wellhead **950**. The platform **600** slides up and down the rail (when positioned at the 90-degree angle) to retrieve drone(s) and elevate them to the height of a lubricator **955** at the top of the wellhead **950**. According to an aspect, the platform **600** may retrieve a magazine **100** from the mobile unit **400** and move the magazine **100** up the rail. Once the magazine is oriented so that it is above or close to the wellhead **950**, drones may be moved from the magazine to the wellhead **950**.

(57) According to an aspect, the platform **600** may be a planar surface. According to an aspect, the platform **600** may include a platform bar **630** that connects to the pair of spaced apart side rails **330** and provide a surface for receiving and securing a “pick and place” manipulator arm **900** (described in further detail hereinbelow) to the platform **600**. According to an aspect, the platform **600** may be configured such that an unrestricted 2-dimensional movement of the manipulator arm **900** in both axes of the horizontal plane is possible, thus enabling an effective and reliable placement of an armed drone to the lubricator.

(58) According to an aspect and as seen in FIG. **13** and FIG. **14** for example, a drone delivery pick-up and deployment arm, also referred to herein as a manipulator arm **900**, is secured to the platform

600. The manipulator arm **900** may retrieve a single drone, or multiple drones at once, from the magazine **100** and deliver them one-by-one to the top of one or several wellheads **950** (in appropriate proximity to the self-erecting launcher assembly **300**). The manipulator arm **900** may have a reach of any amount between about 1 meter to about 10 meters. According to an aspect, the manipulator arm **900** has a reach of about 3 meters to about 5 meters. While FIG. **13** and FIG. **14**, for example, illustrate a single manipulator arm **900**, it is contemplated that two or more manipulator arms **900** may be provided.

(59) According to an aspect, the drone **10** may be charged once it is engaged with the manipulator arm **900**. Once the drone **10** is charged, according to an aspect, the manipulator arm **900** may subject the drone **10** to pre-deployment testing to confirm that the drone **10** is programmed, charged, armed and tested, and will operate according to a given set of parameters. The parameters may be set to confirm that the drone **10** will operate as desired in the wellbore **50**. The parameters may also be set to confirm that the drone selected is of the correct configuration sought to be next dropped into the wellbore **50**. Electrical or signal connections associated with the manipulator arm **900** may perform this testing once the manipulator arm **900** engages the drone **10**.

(60) The programming of a drone **10**, i.e., providing instructions to electronics inside the drone **10**, may be accomplished either previous to or simultaneously with the pre-deployment testing. The details of the programming provided to a particular drone **10** will depend upon the type of drone it is and the details of the job to be performed.

(61) The drone may include an activation pin, latch or ballistic interrupt (not shown) that prevents certain functions from occurring prior to the drone **10** being deployed in wellbore **50**. The ballistic interrupt in an exemplary embodiment of a drone may be positioned to prevent detonation of shaped charges within a perforating drone, such as described in U.S. application Ser. No. 16/451,440 filed Jun. 25, 2019 and Ser. No. 16/542,890 filed Aug. 16, 2019, each of which are commonly owned by DynaEnergetics Europe GmbH. The mechanism for removing or moving the ballistic interrupt may be, without limitation, a hydraulically or electrically powered mechanical engagement for the ballistic interrupt.

(62) For example, in the event that the drone **10** contains explosives or pyrotechnics, it is very important to prevent initiation of these elements prior to dropping the drone **10** into the wellbore. Removal of the ballistic interrupt may be performed by the manipulator arm **900** prior to disposal of the drone **10** into the wellbore **50**. For example, once the platform **600** is on top of the wellhead **950**, the manipulator arm **900** may remove, or move to an armed position, the ballistic interrupt of the drone to ballistically arm the drone **10**. Once the ballistic interrupt has been removed, the drone may be charged.

(63) Alternatively, when the lubricator **955** receives the charged drone, the lubricator **955** removes or moves the ballistic interrupt to the armed position. The drone is ready to deploy/launch into the wellbore and may be “hot” (i.e., activated and when configured as a perforating gun, able to initiate shaped charges).

(64) To be sure, the drone **10** or a plurality of drones in a magazine may be charged when the drone is in the front most shielded position within the receptacle **401**, i.e., a position in the receptacle **401** having metal foam. A user may communicate with the drone at the same time, via a communicative electrical connection on the manipulator arm **900**.

(65) According to an aspect, the manipulator arm **900** is configured to move in X, Y and Z directions to engage and move the drone when the platform **600** is adjacent the magazine and also when the platform **600** is adjacent the receptacle **401** in order to select and place a drone into the lubricator **955**, when the platform **600** is raised to the height of the lubricator **955**. The lubricator **955** may be secured to the wellhead **950**, so that the drone can be delivered from the lubricator **955** to the wellhead **950** and into the wellbore. Once the drone is placed in the lubricator **955**, a mechanism coupled to the lubricator **955** can remove or move the ballistic interrupt positioned in the drone, to arm the drone.

(66) It is contemplated that the platform **600** may move up and down the pair of spaced apart side rails **330** at a preset pace/time. For example, the platform **600** may move up and down the pair of spaced apart side rails **330** in about 5 minutes, or less than 3 minutes. The platform **600** may move down the pair of spaced apart side rails **330** at a different speed than the speed for the platform **600** to move up the pair of spaced apart side rails **330**. For example, the platform **600** may move down the pair of spaced apart side rails **330** in about 2 minutes and may move up the rail in about 3 minutes.

(67) When the platform **600** has moved down the pair of spaced apart side rails **330** to select a drone **10** from a magazine **100**, the drone may have already been charged and/or programmed with instructions for carrying out its wellbore operation while in the magazine **100**. For example, the drone **10** may be charged when a lid of a magazine **100** containing the drone **10** is in a closed position. The lid includes contact(s) for engaging complimentary charging/programming contacts on the drone **10**. In an aspect, electrical contact/communicative communications between the magazine **100** and the drone **10** may be used to test, e.g., the drone control circuitry, electrical connections, and the like. Should a drone **10** fail a diagnostic test, it may be removed from the magazine **100** or moved to another position within the magazine **100**, thereby preventing the manipulator arm **900** from selecting and deploying the drone.

(68) The self-erecting launcher assembly **300** may include an electrical motor (not shown). The electrical motor may be provided on or in the mobile unit **400** or adjacent the mobile unit **400**. The electrical motor is configured to run or power both the platform **600** and arm. For example, the electrical motor may help hoist the elevated platform **600** from the mobile unit **400** to an upper end of the pair of spaced apart side rails **330**. According to an aspect, the electrical motor is an electrical X-motor (i.e., a motor that generates no sparks, etc.) that is safe for use in an explosive atmosphere. The electrical X-motor is particularly helpful because it does not have the potential limitations of hydraulic unit **450s 450**.

(69) In an aspect of the exemplary embodiments, the drone **10** is in communication with a control unit **700** via communicative electrical connection(s) between the drone and the manipulator arm **900**, which is in electrical or wireless communication with the control unit **700**, at all times before the drone is released by the manipulator arm **900** into the lubricator **955**. For purposes of this disclosure, electrical communication may include physical electrically communicative connections or wireless communications, such as radio frequency (RF) transmissions, as are known, and the like. The drone can be retrieved, and the delivery aborted, at any time until placement of the drone into the lubricator **955** takes place.

(70) The control unit **700** may include a controller to monitor, for example, the position of the mobile unit **400** on the track rails **800**. According to an aspect, the controller controls the movement of at least one of the mobile unit **400**, the rotational unit, the conveyor system **430**, the platform **600**, and the manipulator arm **900**. The controller may help to control the process of the deployment of the drones into the wellbore. According to an aspect, the controller plays a supervisory and diagnostic role in monitoring the position and function of at least one of the mobile unit **400**, the rotational unit, the conveyor system **430**, the platform **600**, and the manipulator arm **900**.

(71) For example, the mobile unit **400** travels on track rails **800** secured to a ground surface to access and service multiple wellheads **950** (FIG. 8). The controller may monitor the interface between the mobile unit **400** and the track rails **800**, as well as the condition of the track rails **800** during normal operation of the self-erecting launcher assembly **300**, without interrupting the delivery and deployment of drones into the wellbore **50**. The controller may control an imaging device positioned on the mobile unit **400** or the track rails **800** in order to capture still or video images of the mobile unit **400** in operation. The controller may use the capture images to determine whether irregularities is present in the operation of the self-erecting launcher assembly **300**.

(72) A control room **750** may be provided on the wellsite to handle multiple self-erecting launcher

assemblies **300** at the same time, or a single self-erecting launcher assembly **300** that services multiple wellheads **950**. The control unit **700** may be positioned in the control room **750**. The control room **750** is illustrated in FIG. **15**. According to an aspect, the control room **750** serves as a central unit/area for personnel to monitor the functionality and operation of the self-erecting launcher assembly **300** and each of its components. According to an aspect, the control room **750** is positioned on a skid so that the control room **750** is movable from one location to another location. The control room **750** may alternatively be positioned on the bed of a truck or a trailer that can be pulled by an automobile, such as a truck. The control room **750** can be equipped with an office space (including desks, computer screens, etc.) that is manned by personnel. The control room **750** may be equipped with a generator or may be plugged in to access other power sources that are available at the well site. Such power sources may include a diesel generator that helps to transition the use of the control room **750** at the wellsite.

(73) One or more magazines **100**, such as the magazines **100** described in U.S. Pat. No. 10,605,037 and U.S. Application No. 62/940,480 filed Nov. 26, 2019 and commonly owned by DynaEnergetics Europe GmbH, may be positioned in the magazine loading bay **420** using a forklift **990**.

(74) The forklift **990** may include specially designed adjustable forks **995** that engage the underside of a magazine in order to move the magazine **100** to the magazine loading bay **420**. According to an aspect, the adjustable forks **995** of the forklift **990** may be customizable in order to ensure compatibility with the base frame of the magazine **100**. Most likely the frame size of the magazine will be chosen to be compatible with standard fork-lift dimensions.

(75) Further embodiments of the disclosure may be associated with a method of servicing a plurality of wellheads. The method includes using a self-erecting launcher assembly to deploy a drone into a wellbore. The launcher assembly includes a mobile unit. A forklift or another carrying mechanism may be utilized to move a magazine and position the magazine in a receptacle of the mobile unit. According to an aspect, the magazine moves within an interior of the mobile unit, via a conveyor, for example, to a front portion of the mobile unit. The magazine may be positioned on a platform that is secured to a rail unit. The rail unit, secured to the mobile unit, is moved from a horizontal configuration to a vertical configuration, taking the platform and the magazine with it. The platform may move up the rail unit so that it is at the height of or higher than a lubricator positioned at a wellhead. A manipulator arm, secured to the platform, picks up a drone positioned in the magazine, and places the drone into the lubricator. The manipulator arm may pick up another drone and place it into the lubricator. According to an aspect, the mobile unit may be configured to move between wellheads via wheels or a rotational mechanism. The mobile unit may be positioned on rails that run alongside a plurality of different wellheads, so that the mobile unit can travel along those rails in order to service the different wellheads.

(76) This disclosure, in various embodiments, configurations and aspects, includes components, methods, processes, systems, and/or apparatuses as depicted and described herein, including various embodiments, sub-combinations, and subsets thereof. This disclosure contemplates, in various embodiments, configurations and aspects, the actual or optional use or inclusion of, e.g., components or processes as may be well-known or understood in the art and consistent with this disclosure though not depicted and/or described herein.

(77) The phrases “at least one”, “one or more”, and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “at least one of A, B, or C”, “one or more of A, B, and C”, “one or more of A, B, or C” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together.

(78) In this specification and the claims that follow, reference will be made to a number of terms that have the following meanings. The terms “a” (or “an”) and “the” refer to one or more of that entity, thereby including plural referents unless the context clearly dictates otherwise. As such, the terms “a” (or “an”), “one or more” and “at least one” can be used interchangeably herein.

Furthermore, references to “one embodiment”, “some embodiments”, “an embodiment” and the like are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term such as “about” is not to be limited to the precise value specified. In some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Terms such as “first,” “second,” “upper,” “lower” etc. are used to identify one element from another, and unless otherwise specified are not meant to refer to a particular order or number of elements.

(79) As used herein, the terms “may” and “may be” indicate a possibility of an occurrence within a set of circumstances; a possession of a specified property, characteristic or function; and/or qualify another verb by expressing one or more of an ability, capability, or possibility associated with the qualified verb. Accordingly, usage of “may” and “may be” indicates that a modified term is apparently appropriate, capable, or suitable for an indicated capacity, function, or usage, while taking into account that in some circumstances the modified term may sometimes not be appropriate, capable, or suitable. For example, in some circumstances an event or capacity can be expected, while in other circumstances the event or capacity cannot occur—this distinction is captured by the terms “may” and “may be.”

(80) As used in the claims, the word “comprises” and its grammatical variants logically also subtend and include phrases of varying and differing extent such as for example, but not limited thereto, “consisting essentially of” and “consisting of.” Where necessary, ranges have been supplied, and those ranges are inclusive of all sub-ranges therebetween. It is to be expected that the appended claims should cover variations in the ranges except where this disclosure makes clear the use of a particular range in certain embodiments.

(81) The terms “determine”, “calculate” and “compute,” and variations thereof, as used herein, are used interchangeably and include any type of methodology, process, mathematical operation or technique.

(82) This disclosure is presented for purposes of illustration and description. This disclosure is not limited to the form or forms disclosed herein. In the Detailed Description of this disclosure, for example, various features of some exemplary embodiments are grouped together to representatively describe those and other contemplated embodiments, configurations, and aspects, to the extent that including in this disclosure a description of every potential embodiment, variant, and combination of features is not feasible. Thus, the features of the disclosed embodiments, configurations, and aspects may be combined in alternate embodiments, configurations, and aspects not expressly discussed above. For example, the features recited in the following claims lie in less than all features of a single disclosed embodiment, configuration, or aspect. Thus, the following claims are hereby incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment of this disclosure.

(83) Advances in science and technology may provide variations that are not necessarily express in the terminology of this disclosure although the claims would not necessarily exclude these variations.

Claims

1. A self-erecting launcher assembly comprising: a mobile unit comprising a receptacle; a magazine including a plurality of drones positioned in the receptacle; a rail unit secured to the mobile unit, the rail unit movable between a closed position and an elevated position; and a platform configured to move the magazine or the plurality of drones from the mobile unit to at least one wellhead.
2. The self-erecting launcher assembly of claim 1, wherein the mobile unit further comprises a

magazine loading bay in communication with the receptacle.

3. The self-erecting launcher assembly of claim 1, wherein the mobile unit further comprises: a conveyor system in the receptacle; and a magazine retrieval zone in communication with the receptacle, wherein the conveyor system is configured to move the magazine from a magazine loading bay to the magazine retrieval zone.

4. The self-erecting launcher assembly of claim 3, wherein the magazine is configured to move within the receptacle to the magazine retrieval zone via the conveyor system.

5. The self-erecting launcher assembly of claim 1, further comprising: a ballistic limiter configured to prevent or limit ballistic transfer from a first drone to a second drone positioned in the magazine in the event of detonation of an explosive component of the first drone.

6. The self-erecting launcher assembly of claim 1, wherein the rail unit further comprises: a pair of spaced apart side rails, wherein a first rail of the pair of spaced apart side rails includes a first end pivotably connected to the mobile unit, and a second end spaced apart from the first end, and a second rail of the pair of spaced apart side rails includes a first end pivotably connected to the mobile unit, and a second end spaced apart from the first end.

7. The self-erecting launcher assembly of claim 6, further comprising: a pair of telescoping arms, wherein a first telescoping arm of the pair of telescoping arms is configured to move a first rail of the pair of spaced apart side rails, and a second telescoping arm of the pair of telescoping arms is configured to move a second rail of the pair of spaced apart side rails.

8. The self-erecting launcher assembly of claim 7, wherein the first telescoping arm is connected to the first rail at a position between the first end and the second end of the first rail, and the second telescoping arm is connected to the second rail at a position between the first end and the second end of the second rail.

9. The self-erecting launcher assembly of claim 1, further comprising: a manipulator arm secured to the platform, wherein the manipulator arm is configured to select a drone of the plurality of drones and delivers the selected drone into a lubricator.

10. The self-erecting launcher assembly of claim 9, further comprising: a control unit configured to communicate with at least one of the mobile unit, the magazine, the plurality of drones, the rail unit, the platform, the manipulator arm, and an electrical motor.

11. A wellbore deployment system comprising: a plurality of self-erecting launcher assemblies, each self-erecting launcher assembly of the plurality of self-erecting launcher assemblies comprising: a mobile unit comprising a receptacle; a magazine including a plurality of drones positioned in the receptacle; a rail unit secured to the mobile unit, the rail unit movable between a closed position and an elevated position; a platform configured to move the magazine or the plurality of drones from the mobile unit to at least one wellhead; and a control unit configured to manage functionality and operation of the plurality of self-erecting launcher assemblies.

12. The wellbore deployment system of claim 11, wherein the mobile unit comprises: at least one side wall having an upper portion and a lower portion; a roof connected to the upper portion of the at least one side wall; and a floor connected to the lower portion of the at least one side wall, wherein the receptacle is bounded by the at least one side wall, the roof and the floor.

13. The wellbore deployment system of claim 12, wherein the mobile unit further comprises a magazine loading bay in communication with the receptacle.

14. The wellbore deployment system of claim 13, wherein the magazine loading bay is positioned in the at least one side wall.

15. The wellbore deployment system of any one of claim 11, wherein the rail unit further comprises: a pair of spaced apart side rails, wherein a first rail of the pair of spaced apart side rails includes a first end pivotably connected to the mobile unit, and a second end spaced apart from the first end, and a second rail of the pair of spaced apart side rails includes a first end pivotably connected to the mobile unit, and a second end spaced apart from the first end.

16. A wellbore deployment system comprising: a self-erecting launcher assembly comprising: a

mobile unit comprising a receptacle; a magazine positioned in the receptacle, the magazine including a plurality of drones; a rail unit secured to the mobile unit, the rail unit movable between a closed position and an elevated position; and a platform configured to move the magazine or the plurality of drones from the mobile unit to at least one wellhead; and a location adjuster configured to move the self-erecting launcher assembly from a first wellhead to additional wellheads.

17. The wellbore deployment system of claim 16, wherein the location adjuster comprises at least one of: a plurality of wheels secured to the mobile unit; and a track rail.

18. The wellbore deployment system of claim 16, wherein the location adjuster comprises: a rotational unit, wherein the mobile unit is positioned on the rotational unit, and the mobile unit is movable between a first wellhead and additional wellheads.

19. The wellbore deployment system of claim 16, wherein the location adjuster comprises: a scissor lift configured to move the mobile unit in an upward direction to the height of the wellhead.

20. The wellbore deployment system of claim 16, wherein the rail unit further comprises: a pair of spaced apart side rails, wherein a first rail of the pair of spaced apart side rails includes a first end pivotably connected to the mobile unit, and a second end spaced apart from the first end, and a second rail of the pair of spaced apart side rails includes a first end pivotably connected to the mobile unit, and a second end spaced apart from the first end.
